Digital Accessibility in the AI era: Promise, Pitfalls, and Progress

From legal frameworks to AI-powered tools

Pesaresi Seminar Series
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What is Digital Accessibility?

01

Physical accessibility ensures that environments, products, and services are usable by people with visual, hearing, motor, or cognitive disabilities.
Digital accessibility is the same principle, applied to digital spaces and content. **Digital content must be usable by everyone**.



WHAT it covers

- Web content
- Mobile applications
- Desktop software & OS
- Digital documents (PDF, EPUB)
- Multimedia & streaming content



WHY it matters

3 Pillars*

- **Inclusion:** fostering independence, dignity, and equal participation
- **Profit:** reach more customers
- **Compliance**: Legal requirement in most countries

And many other reasons

- Improves UX for all users
- Corporate social responsibility
- Brand Identity
- ...

*Byrne-Haber's https://www.sheribyrnehaber.com/

WHO it affects?

MISCONCEPTION 1: DISABILITIES ARE RARE

- ~ 5% percent of people are born with disabilities.
- *Increased to ~17% due to disease, accidents and age.*

*More than **1.3 billion people*** **worldwide (1/6)** live with disabilities, according to the World Health Organisation*.

REALITY: Factoring in temporary and situational conditions, accessibility is something **almost everyone** relies on at some point.

The Disability Matrix

	Temporary	Situational	Permanent
Visual	Pregnancy-related vision loss	Driving a car (eyes on the road)	Blindness
Hearing	Ear infection	Loud concert or noisy bar	Deafness
Motor	Broken arm	Holding a baby	Missing limb
Cognitive	Sleep-deprived parent	Anxiety due an upcoming event	Dyslexia/ ADHA

* World Health Organization. 2024. Disability. https://www.who.int/health-topics/disability

WHAT does it affect?

MISCONCEPTION 2: ASSISTIVE AND ACCESSIBLE TECHNOLOGIES ARE A NICHE MARKET

Universal Design Benefits Everyone*: Innovations like audiobooks, OCR and captioning, originally for disabled users, now enhance usability for all.*

REALITY: What is initially designed for a niche of users can be extended to the masses, multiplying its impact far beyond its original scope. **"Solve for one, extend to many"** principle**.**
Assistive technologies often become the foundation of mainstream innovations. This makes digital accessibility a **clear business strategy**.

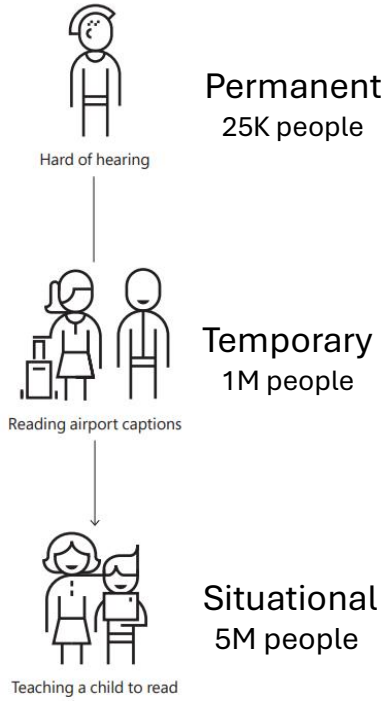


Image credits: Microsoft. Inclusive Design Toolkit Manual

Regulatory and Technical Frameworks

WCAG — The Global Standard, The Technical Guide

- Web Content Accessibility Guidelines by W3C
- WCAG 2.1 (2018): **current baseline**
- WCAG 2.2 (2023): new criteria for cognitive & mobile
- WCAG 3.0: in development — outcome-based model

The guideline has been regularly revised to keep pace with **evolving technology** and better serve people with a **wider range** of disabilities.

The LAWS

EU European Accessibility Act (EAA)

Directive 2019/882 — **mandatory** for products & services in the EU market by **June 2025**. Covers e-commerce, banking, transport, e-books.

US Section 508 (USA)

Federal agencies must make ICT accessible. ADA extends requirements to private companies. Enforcement through DOJ and civil litigation.

🌍 Other

Italy: **Legge Stanca (2004)** + recent EAA transposition. France: RGAA (2019). Canada: ACA (2019). Many nations adopt WCAG as baseline.

Technical Framework WCAG 2.1

WCAG — The Global Standard, The Technical Guide

Several layers of guidance:

- **4 Principles**: POUR
- **13 Guidelines** : not directly testable
- **78 Success Criteria** (conformance level A, AA, AAA): testable
- Other sufficient and advisory **Techniques**

The Structure of the Framework

Principles	Guidelines	Success Criteria A	AA	AAA
1. Perceivable	1.1 Text Alternatives	1.1.1		
	1.2 Time-based Media	1.2.1 – 1.2.3	1.2.4 – 1.2.5	1.2.6 – 1.2.9
	1.3 Adaptable	1.3.1 – 1.3.3		
	1.4 Distinguishable	1.4.1 – 1.4.2	1.4.3 – 1.4.5	1.4.6 – 1.4.9
2. Operable	2.1 Keyboard Accessible	2.1.1 – 2.1.2		2.1.3
	2.2 Enough Time	2.2.1 – 2.2.2		2.2.3 – 2.2.5
	2.3 Seizures	2.3.1		2.3.2
	2.4 Navigable	2.4.1 – 2.4.4	2.4.5 – 2.4.7	2.4.8 – 2.4.10
3. Understandable	3.1 Readable	3.1.1	3.1.2	3.1.3 – 3.1.6
	3.2 Predictable	3.2.1 – 3.2.2	3.2.3 – 3.2.4	3.2.5
	3.3 Input Assistance	3.3.1 – 3.3.2	3.3.3 – 3.3.4	3.3.5 – 3.3.6
4. Robust	4.1 Compatible	4.1.1 – 4.1.2		

Evaluation Tools

Manual & Assistive Technology

Screen Readers

NVDA (free, Windows), JAWS (commercial), VoiceOver (Mac/iOS), TalkBack (Android).

Keyboard Navigation Testing

Tab order, focus indicators, skip links — tested manually without a mouse.

Colour Contrast Analysers

Colour Contrast Analyser (TPGi), Contrast Checker (WebAIM) verify WCAG ratios.

Expert Audits

Certified professionals (CPACC, WAS) test against WCAG using structured test plans.

WCAG Success Criteria **are defined in conjunction with the tools** used by people with disabilities. Expert evaluators must use the same tools and methods to perform an effective evaluation. (e.g. image alternative-text is read by screen readers; keyboard-only navigation to assess trap)

Evaluation Tools

Automated Scanners (the most used ones)

axe DevTools

Browser extension & API by Deque. Industry-standard; zero false positives policy. https://www.deque.com/axe/

WAVE

WebAIM's visual in-page annotator. Great for education and quick audits. https://wave.webaim.org/

Lighthouse

Built into Google Chrome DevTools. Scores pages and integrates into CI/CD pipelines. link

QualWeb

Comes from different research projects. Automatically check a webpage against a set of WCAG 2.1 Techniques and ACT Rules. link

Mauve ++

Free platforms from **CNR (Pisa)** for large-scale automated scanning across full websites and PDF contents. https://mauve.isti.cnr.it/

⚠ **94.8%** failure rate in the **top 1 million** home pages with an average of **51 errors** per page (**4.1%** home page elements have an accessibility error). (2025 report)*

Key Advice: Current automated tools detect **~30–40% of WCAG issues**. The numbers above are extremely optimistic (these tools can only perform partial checks e.g. for colour contrast).

On the other hand: A website can achieve 99% compliance but still be **entirely unusable** for certain types of users.

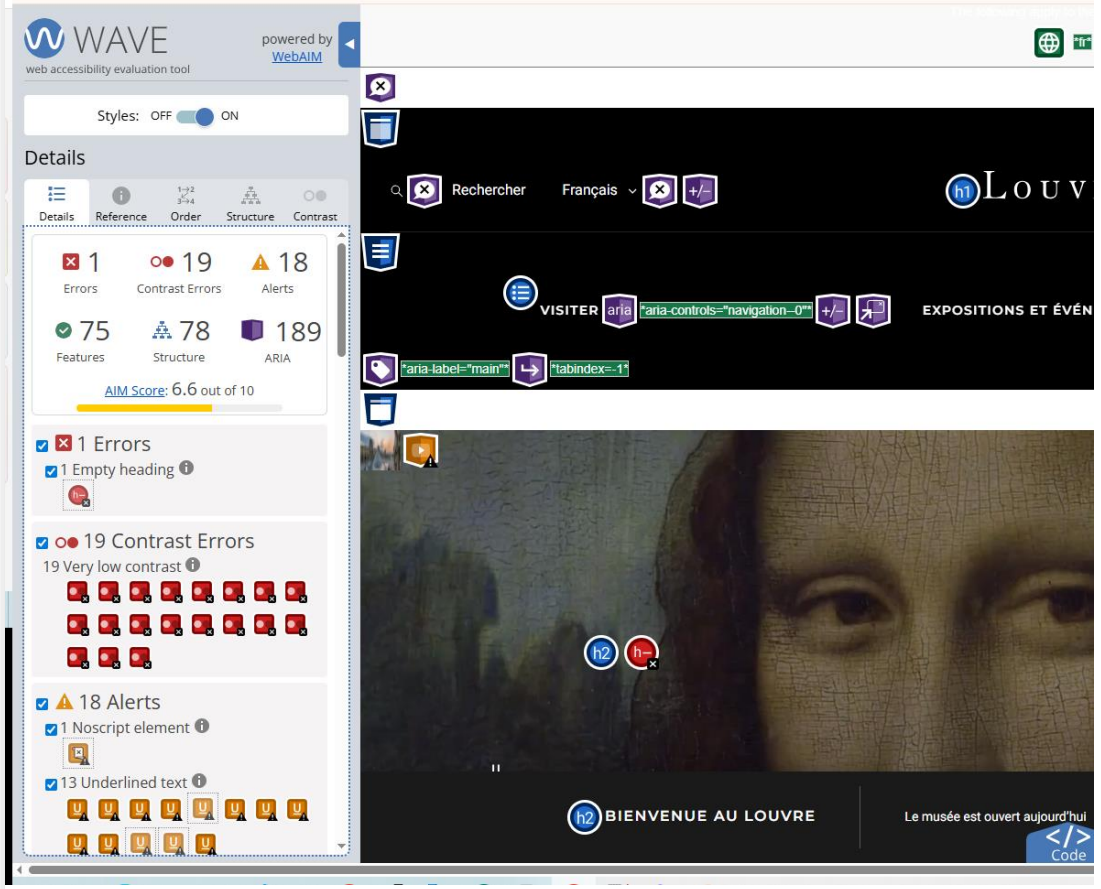
Expert audit and test with people with disabilities remains essential.

*WebAIM Million Report

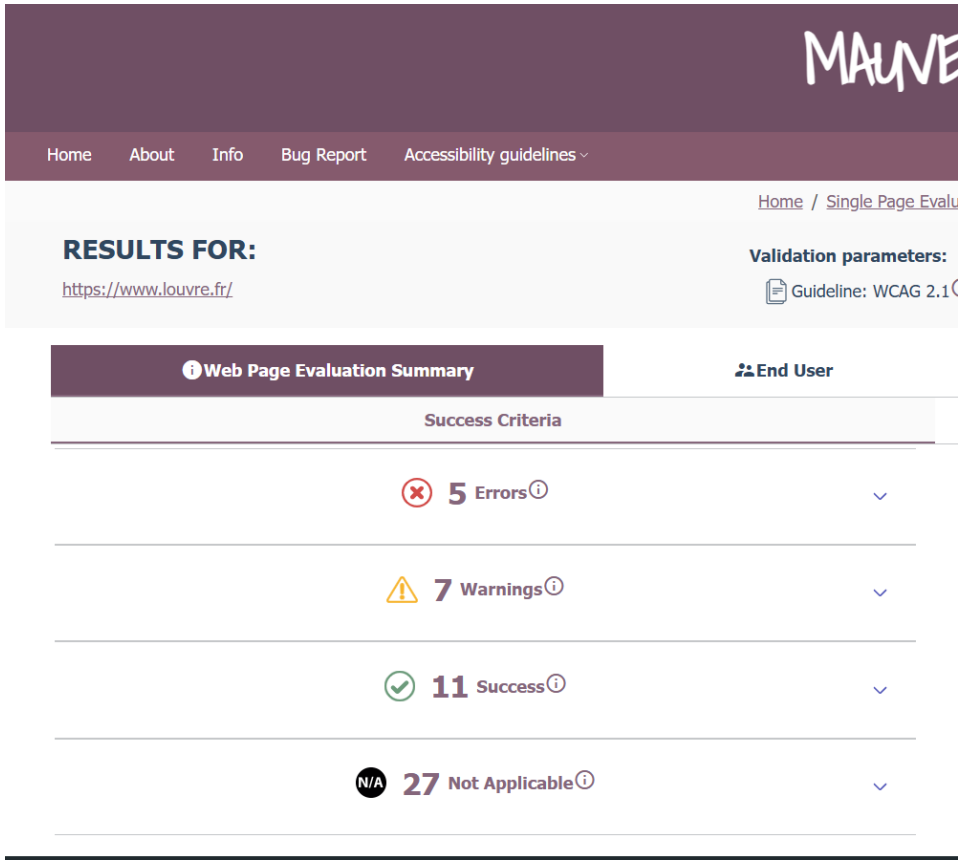
Evaluation Tools Output Comparison:

Louvre website https://www.louvre.fr

Wave Outputs



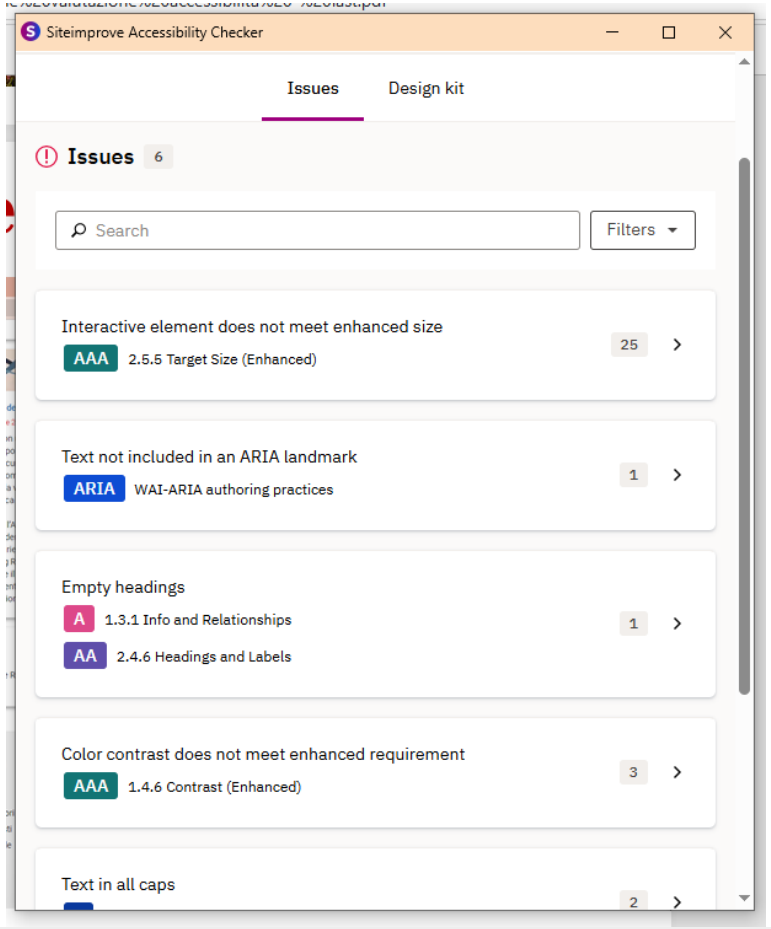
Mauve++ Outputs



Evaluation Tools Output Comparison:

Louvre website https://www.louvre.fr

Siteimprove Outputs



Comments

Different Accessibility Tools Report Different Results

- **Detection method varies**: real in-browser rendering (Wave, axe) VS static DOM analysis (Siteimprove)
- **Severity classification differs**: the same issue may be a critical error in one tool and a warning in another
- **Guideline Interpretation** : WCAG success **criteria are intentionally technology-agnostic;** written as principles, not precise technical rules, they leaves room for different "correct" implementations and readings. Tools encode one **interpretation** (like human experts ...)

Takeaway: No single tool provides the full picture; combining multiple tools with manual testing is the only way to build a reliable accessibility baseline.

Evaluation Tools in the Pre-AI Era : What was missing?

1.	**Perceivable**
1.1	Text Alternatives
1.1.1	Non-text Content
1.2	Time-based Media
1.2.1	Audio-only and Video-only (Prerecorded)
1.2.2	Captions (Prerecorded)
1.2.3	Audio Description or Media Alternative (Prerecorded)
1.2.4	Captions (Live)
1.2.5	Audio Description (Prerecorded)
1.2.6	Sign Language (Prerecorded)
1.2.7	Extended Audio Description (Prerecorded)
1.2.8	Media Alternative (Prerecorded)
1.2.9	Audio-only (Live)
1.3	Adaptable
1.3.1	Info and Relationships
1.3.2	Meaningful Sequence
1.3.3	Sensory Characteristics
1.3.4	Orientation
1.3.5	Identify Input Purpose
1.3.6	Identify Purpose
1.4	Distinguishable
1.4.1	Use of Color
1.4.2	Audio Control
1.4.3	Contrast (Minimum)
1.4.4	Resize Text
1.4.5	Images of Text
1.4.6	Contrast (Enhanced)
1.4.7	Low or No Background Audio
1.4.8	Visual Presentation
1.4.9	Images of Text (No Exception)
1.4.10	Reflow
1.4.11	Non-text Contrast
1.4.12	Text Spacing
1.4.13	Content on Hover or Focus

2.	**Operable**
2.1	Keyboard Accessible
2.1.1	Keyboard
2.1.2	No Keyboard Trap
2.1.3	Keyboard (No Exception)
2.1.4	Character Key Shortcuts
2.2	Enough Time
2.2.1	Timing Adjustable
2.2.2	Pause, Stop, Hide
2.2.3	No Timing
2.2.4	Interruptions
2.2.5	Re-authenticating
2.2.6	Timeouts
2.3	Seizures and Physical Reactions
2.3.1	Three Flashes or Below Threshold
2.3.2	Three Flashes
2.3.3	Animation from Interactions
2.4	Navigable
2.4.1	Bypass Blocks
2.4.2	Page Titled
2.4.3	Focus Order
2.4.4	Link Purpose (In Context)
2.4.5	Multiple Ways
2.4.6	Headings and Labels
2.4.7	Focus Visible
2.4.8	Location
2.4.9	Link Purpose (Link Only)
2.4.10	Section Headings
2.5	Input Modalities
2.5.1	Pointer Gestures
2.5.2	Pointer Cancellation
2.5.3	Label in Name
2.5.4	Motion Actuation
2.5.5	Target Size
2.5.6	Concurrent Input Mechanisms

3.	**Understandable**
3.1	Readable
3.1.1	Language of Page
3.1.2	Language of Parts
3.1.3	Unusual Words
3.1.4	Abbreviations
3.1.5	Reading Level
3.1.6	Pronunciation
3.2	Predictable
3.2.1	On Focus
3.2.2	On Input
3.2.3	Consistent Navigation
3.2.4	Consistent Identification
3.2.5	Change on Request
3.3	Input Assistance
3.3.1	Error Identification
3.3.2	Labels or Instructions
3.3.3	Error Suggestion
3.3.4	Error Prevention (Legal, Financial, Data)
3.3.5	Help
3.3.6	Error Prevention (All)

4.	**Robust**
4.1	Compatible
4.1.1	Parsing
4.1.2	Name, Role, Value
4.1.3	Status Messages

Current accessibility evaluation tools have **significant blind spots**: they cannot assess semantics, UI interaction, alternative content, or cognitive load. They are **Rule-Based Systems.**

No reasoning, no context awareness: the tool doesn't "understand" the content; it pattern-matches against its predefined ruleset

What rule-based tools do well	**What they struggle with**
Missing alt text, labels, roles	Whether alt text is *descriptive*
Color contrast ratios	Whether content is *understandable*
Duplicate IDs, invalid ARIA	Complex keyboard navigation flows
Structural markup issues	Cognitive accessibility barriers

Can AI help in this direction?

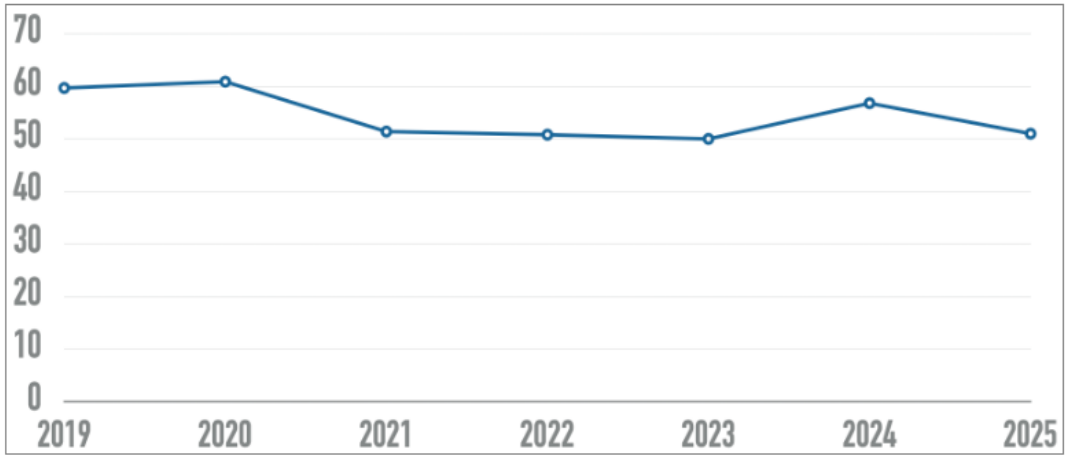
Can AI help in this direction?

What the Research on the topic tells us:

Currently, AI solutions show:

- **Limited** ability **to generate** accessibility-compliant code [2],[3],[8]
- **Limited** effectiveness in accessibility-focused **code remediation** [6],[7],[9]
- **Good-level** performance **in identifying** accessibility issues in real websites (at least for a subset of criteria) [4],[5]

Average number of detected errors per home page over time.



Most common WCAG 2 failures.

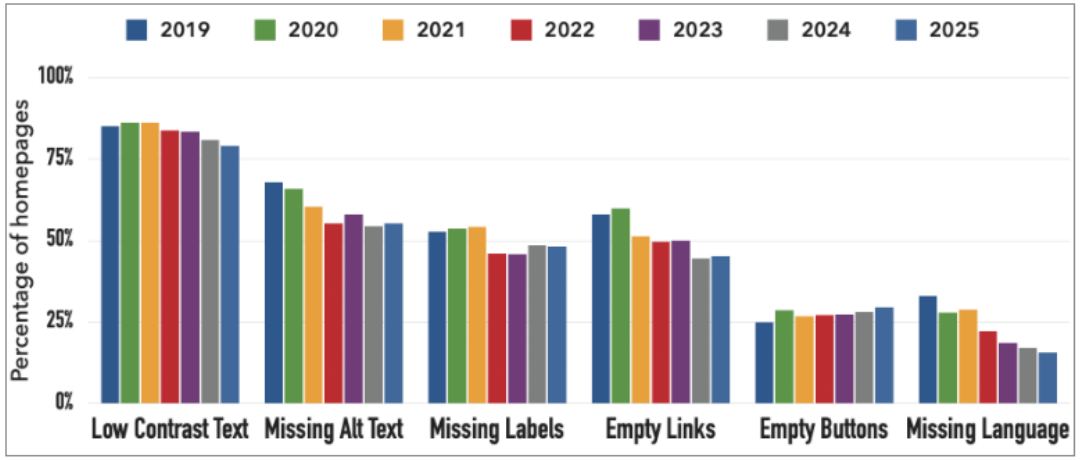
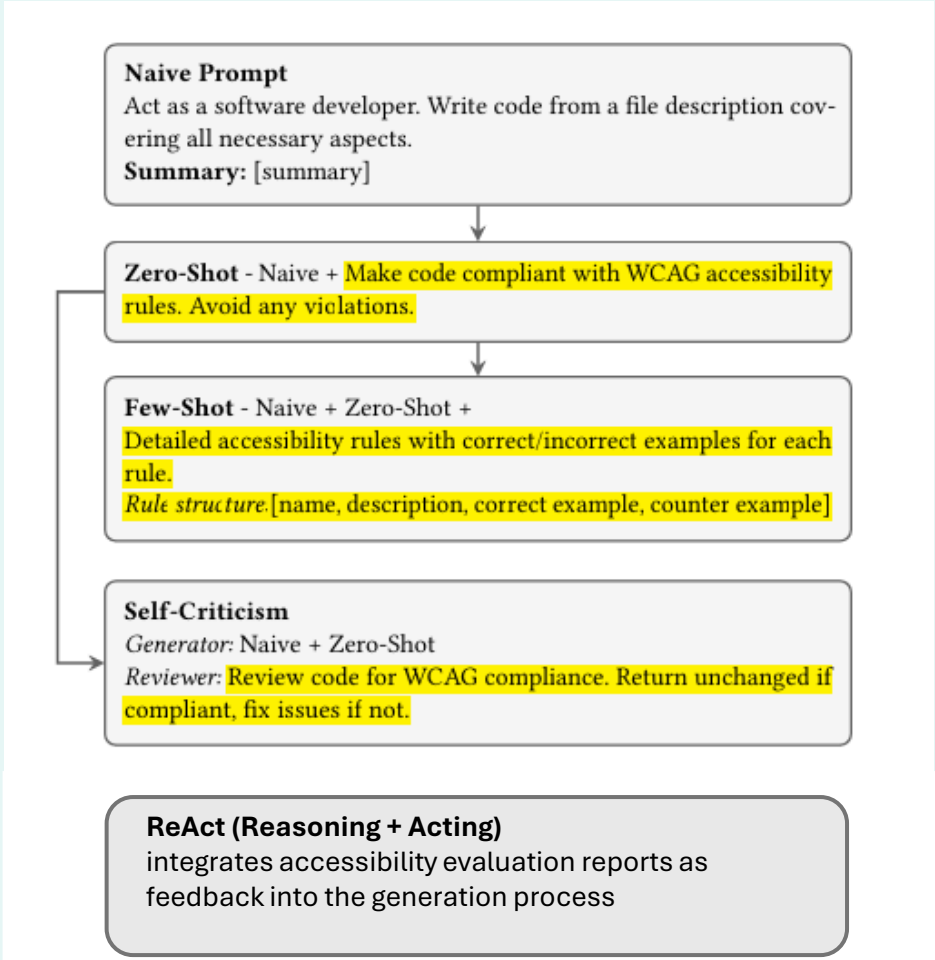


Image credits: WebAIM Million Report

Approaches to code accessibility-compliant generation

Human or LLM? A Comparative Study on Accessible Code Generation Capability [8].



Inaccessibility Rate Computed Using QualWeb for GPT-4o and Qwen2.5-Coder

H	Naive		Zero-Shot		Few-Shot		Self-Criticism		FeedA11y	
	G	Q	G	Q	G	Q	G	Q	G	Q
0.166	0.112	0.139	0.152	0.163	0.174	0.156	0.251	0.141	0.163	0.163
0.021	0.028	0.031	0.024	0.040	0.055	0.044	0.034	0.054	0.025	0.042
0.061	0.070	0.062	0.070	0.069	0.082	0.077	0.082	0.069	0.082	0.065
0.021	0.018	0.025	0.041	0.013	0.026	0.023	0.022	0.092	0.025	0.019
0.144	0.115	0.172	0.139	0.149	0.142	0.136	0.110	0.210	0.169	0.169
0.179	0.028	0.027	0.039	0.032	0.028	0.036	0.028	0.029	0.031	0.031
0.057	0.088	0.091	0.071	0.084	0.060	0.056	0.073	0.091	0.044	0.067
0.266	0.261	0.235	0.263	0.254	0.245	0.263	0.249	0.237	0.250	0.192
0.210	0.236	0.227	0.218	0.256	0.263	0.209	0.209	0.205	0.223	0.223
0.121	0.097	0.121	0.098	0.153	0.102	0.108	0.067	0.087	0.104	0.101
0.125	**0.105**	0.113	0.112	0.121	0.118	0.111	0.112	0.122	0.112	**0.107**

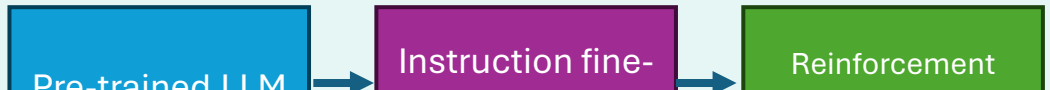
H: Human, **G**: GPT-4o, **Q**: Qwen2.5-Coder, **Naive**: Naive Code Generation

Despite the use of large frontiers models, advanced prompting, and **inference scaling techniques** (higher computational costs), the **inaccessibility rate*** (lower is better) remains **almost constant** and comparable to the human one. Some problems have been fixed, but new ones have been introduced.

*Inaccessibility rate H: the number of elements with accessibility violations over the total number of elements that are prone to accessibility violations (DOM elements).
These numbers are calculated using QualWeb evaluator tool.

Approaches to code accessibility-compliant generation

A11YN: ALIGNING LLMS FOR ACCESSIBLE WEB UI CODE GENERATION *[7]*



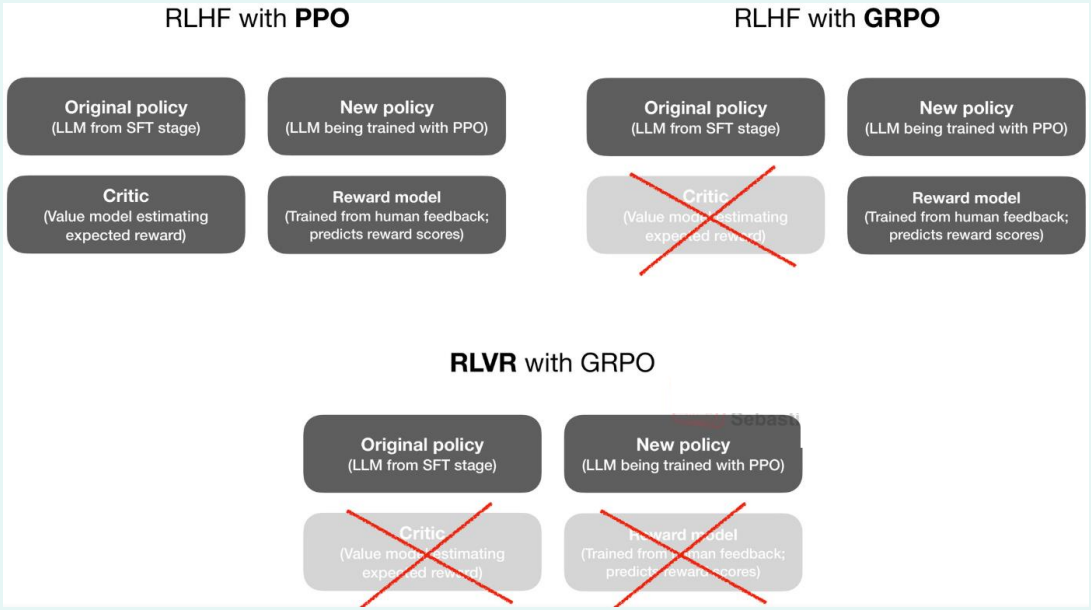


*Image Credits https://magazine.sebastianraschka.com/p/technical-deepseek?utm_source=publication-search

RLHF is a common step* in LLM development that aligns the LLM with human preferences.

Traditional RLHF with PPO uses both a **reward model** (trained on human preferences) and a **critic** (value model) to guide learning.
GRPO eliminates the critic model.
RLVR with GRPO also remove the reward model, relying instead on **verifiable rewards** from external tools (calculators, compilers, etc.). It is no longer based on human preferences. It focuses on reasoning capabilities.

Reinforcement Learning with Human Feedback (RLHF)
Proximal Policy Optimization (PPO)
Group Relative Policy Optimization (GRPO)
Reinforcement Learning with Verifiable Rewards (RLVR) - DeepSeek R1

*InstructGPT paper, https://arxiv.org/abs/2203.02155

Approaches to code accessibility-compliant generation

A11YN: ALIGNING LLMS FOR ACCESSIBLE WEB UI CODE GENERATION *[7]*

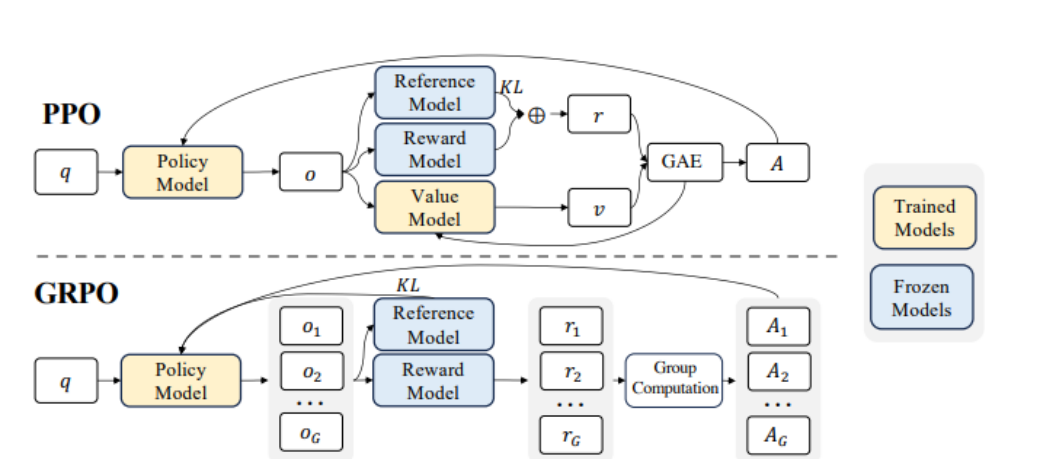
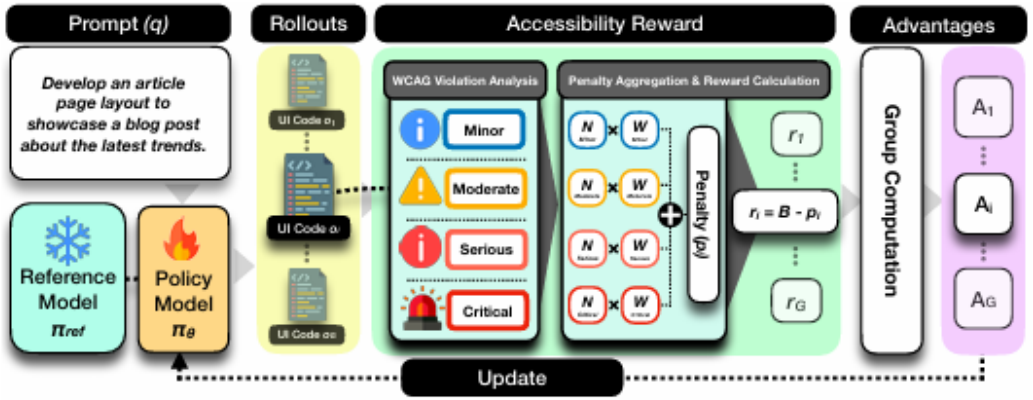


*Image Credits: Deep Seek R1 paper

The idea is to **get the reward signal from a WCAG validator** (Axe)



Weighted Violation Score

$$\text{WVS} = \lambda_{\text{Minor}} \cdot N_{\text{Minor}} + \lambda_{\text{Moderate}} \cdot N_{\text{Moderate}} + \lambda_{\text{Serious}} \cdot N_{\text{Serious}} + \lambda_{\text{Critical}} \cdot N_{\text{Critical}}$$

$$\text{Inaccessibility Rate} = \frac{\text{WVS}}{\text{No. of Total DOM Elements}}$$

Model	Average Violated DOM Counts (↓)				WVS (↓)	IR (↓)
	Minor	Moderate	Serious	Critical		
Qwen2.5-Coder-7B-Instruct	1 (±1)	1149 (±36)	978 (±48)	40 (±0)	5392 (±35)	0.38 (±0.0)
+ Feeda11y	11 (±1)	461 (±37)	841 (±33)	30 (±5)	3576 (±64)	0.21 (±0.0)
Qwen2.5-Coder-14B-Instruct	3 (±2)	846 (±35)	1491 (±32)	49 (±8)	6365 (±158)	0.43 (±0.0)
GPT-4.1	45 (±5)	1925 (±34)	1424 (±21)	105 (±7)	8588 (±100)	0.27 (±0.0)
Claude Sonnet 4	2 (±3)	3388 (±81)	1435 (±14)	282 (±10)	12210 (±117)	0.29 (±0.0)
A11yn (Ours)	**0** (±0)	**231** (±16)	**481** (±23)	**24** (±3)	**1918** (±65)	**0.15** (±0.0)

Despite the improvements, we are still far from fully accessible content (15% of DOM elements contain accessibility errors). Furthermore, these numbers are calculated using rule-based validators that cover a small fraction of real accessibility errors.

Two Limitations:

1. The rewards come from a rule-based accessibility validator (with the well-known limitations).
2. Human feedbacks are not considered

Can AI help in this direction?

Why doesn't AI seem to be very helpful?

LLMs are trained predominantly on non-accessible web content, so the outputs they generate tend to replicate the same accessibility gaps.

"Garbage In, Garbage Out" principle.

What do we need to improve the effective use of AI?

Achieving reliable WCAG compliance requires **shifting from 'web-scale' to 'expert-scale'** datasets curated by **accessibility experts** and validated by **users with disabilities**.

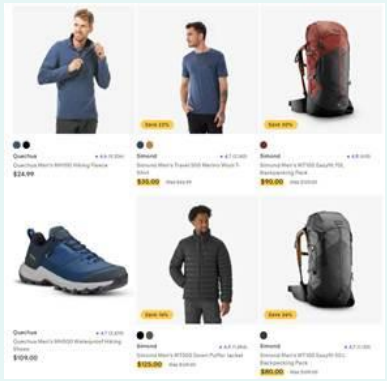
True accessibility in AI-generated content isn't found in more data, but in better data used to train the model.

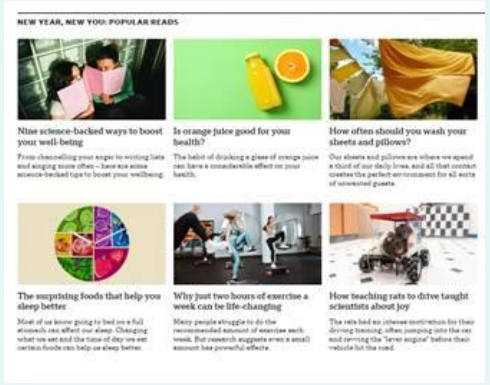
The Case Study: LLM alt-text validation and generation

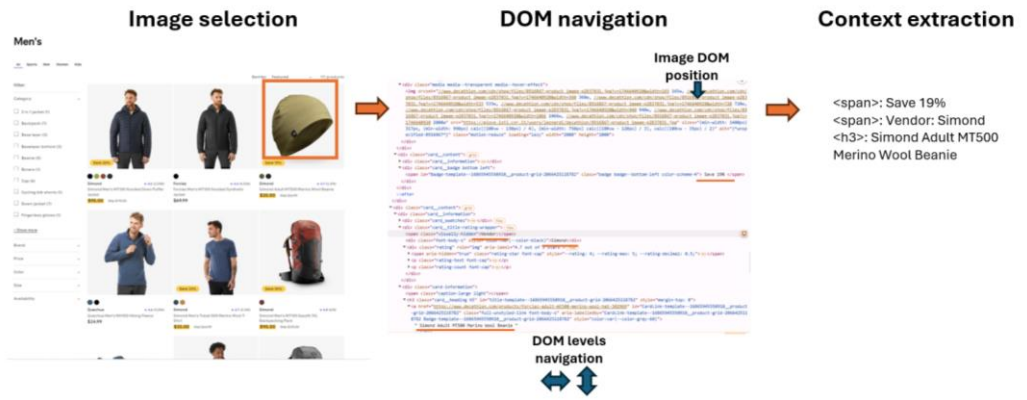
Q1: Is it possible to use LLM as a large-scale screening tool capable of identifying alt-text issues?

G94 technique
https://www.w3.org/WAI/WCAG22/Techniques/general/G94
“The objective of this technique is to create a text alternative that serves the same purpose and presents the same information as the original non-text content…”

The image alone is not enough; context is required.
This is a multimodal task, vision-language model required (GPT-4o).







The Task:

- 16 human validators (with experience in accessibility)
- 5 real website e-commerce (eBay, Amazon, Etsy, Decathlon, Nike)
- ~ 500 image alt-texts were evaluated (1-5 scale) and generated (from both humans and LLM)
- Humans also evaluated LLM-generated alt-texts

Leonardi N., Manca M., Paternò F. *Leveraging Large Language Models for Alt-Text Evaluation in E-Commerce: A Data-Driven Study*.
Accepted at AVI (Advanced Visual Interfaces) 2026 conference

The Case Study: LLM alt-text validation and generation

Analysis Results

Inter-User agreement
Spearman coefficient (0.48)
Cohen's Kappa (0.31)

<

User-LLM agreement
Spearman coefficient (0.59)
Cohen's Kappa (0.55)

<

LLM intra-agreement
Spearman coefficient (0.66)
Cohen's Kappa (0.65)

LLM as a classifier (good VS bad alt-text)
Accuracy: 93%
Precision: 94.7%
Recall: 95.5%
F1 Score: 95.1%

With a well-crafted prompt and an accurate context extraction (multimodal RAG):
The LLM is a more consistent alt-text evaluator than individual humans. The model could be confidently deployed at scale for automated quality screening, flagging inadequate alt-texts for human review.

Leonardi N., Manca M., Paternò F. *Leveraging Large Language Models for Alt-Text Evaluation in E-Commerce: A Data-Driven Study*.
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The Case Study: LLM alt-text validation and generation

Q2: Can LLM generate valid and descriptive alternative texts that meet accessibility guidelines?

The Observed Results
No improvement: 36.64%
Positive improvement: 44.53%
Negative variation: 18.83%

But the results are still distant from the real users' desiderata. How do the alternative text proposals for Human and LLM differ?

- LLMs produce longer texts with higher reading complexity.
- Humans and LLMs describe images with comparable semantic content but divergent surface-level expression. Convergent understanding despite stylistic differences (Lexical, Semantic, BERTscore similarities).
- LLMs are more visually driven in their assessments, whereas humans rely more on contextual information (Kolmogorov-Smirnov Statistic, Wassertein Distance, Jensen-Shannon Divergence, Pearson tests).

Human and LLM proposed alt-text readability descriptive statistics

	Human Flesh Reading Ease index	LLM Flesh Reading Ease Index	Human Gunning Fog index	LLM Gunning Fog index
mean	66.07	59.82	16.27	17.96
std	25.3	20.38	7.32	6.1
min	-96.26	-31.35	0	0.8
25%	53.13	47.06	11.6	13.2
50%	67.14	61.21	16.67	18.56
75%	80.14	73.64	21.35	22
max	129.05	114.09	41.2	35.73

The Flesch Reading Ease Index (0-100); higher scores mean easier reading
The Gunning Fog Index estimates the years of formal education needed to understand text; lower scores mean easier reading

Human and LLM proposed alt-text CLIP scores descriptive statistics.

	alt-text Human	alt-text LLM (context+image)	alt-text LLM (only context)	alt-text LLM (only image)
mean	24.72	26.11	24.92	26.07
std	5.34	4.13	5.35	4.39
min	0*	10.38	5.36	12.74
25%	21.99	23.72	22.17	23.23
50%	25.61	26.53	25.91	26.52
75%	28.06	28.64	28.27	29.10
max	38.11	38.11	39.11	38.34

*0 means no alt-text provided, considered a decorative image

Future Work: In progress

Use an open-weight small vision-language model (<4B parameters)

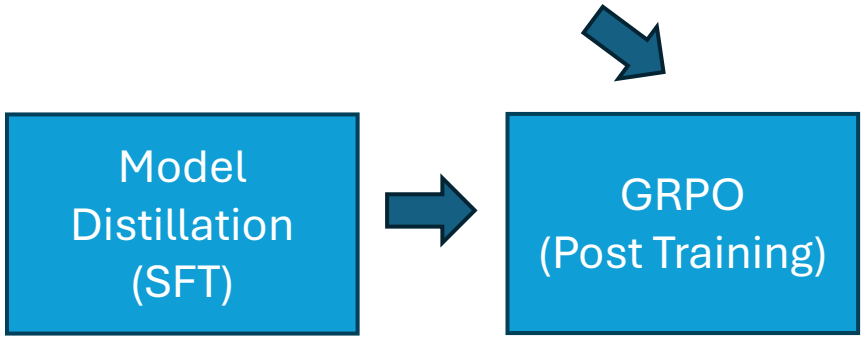
PROS

- Can run locally on a «consumer hardware» reducing operative costs
- Privacy Preservation
- Reduced Latency
- Minimised Energy Consumption and Carbon Emissions
- Full control over model weights, thus we can fine-tune the model

CONS

- Requires in-house infrastructure and ongoing maintenance
- Lower performance out of the box

There are potentially verifiable human signals that can be used as rewards



Supervised Finetuning From Big Model Synthetic data

Aligned with rewards signals coming from **human preferences**

The expectation is an improvement in performance in both alt-text assessment and alt-text generation quality.
The approach can be extended to other success criteria verification and, consequently, to **accessibility-compliant generation solutions.**

Key Takeaways

01 Digital Accessibility matters to a lot of people in the World.

02 Digital Accessibility is a legal requirement and a fundamental right. Is not optional.

03 Automated tools (pre-AI and AI eras) are essential, but with known issues and limitations.

04 Simply more data is not the right path; we need better data

05 Human expertise + user testing with disabled people remain irreplaceable.

Key Resources

W3C WCAG Guidelines
w3.org/WAI/WCAG21

Mauve++
mauve.isti.cnr

WebAIM
webaim.org/projects/million

EU Accessibility Act
ec.europa.eu/accessibility

A11Y Project
a11yproject.com

Microsoft Inclusive Design
inclusive.microsoft.design

References

[1] *Microsoft. Inclusive Design Toolkit Manual. Available as a PDF via Microsoft's download centre:* *https://download.microsoft.com/download/b/0/d/b0d4bf87-09ce-4417-8f28-d60703d672ed/inclusive_toolkit_manual_final.pdf*

[2] Wajdi Aljedaani, Abdulrahman Habib, Ahmed Aljohani, Marcelo Eler, and Yunhe Feng. 2024. Does ChatGPT Generate Accessible Code? Investigating Accessibility Challenges in LLM Generated Source Code. In Proceedings of the 21nd International Web for All Conference. 165–176.

[3] Alexandra-Elena Guriţa and Radu-Daniel Vatavu. 2025. When LLM-Generated Code Perpetuates User Interface Accessibility Barriers, How Can We Break the Cycle. In Proceedings of the 22nd International Web for All Conference (W4A'25).

[4] Ziyao He, Syed Fatiul Huq, and Sam Malek. 2024. Enhancing Web Accessibility: Automated Detection of Issues with Generative AI. https://github.com/seal hub/GenA11y

[5] Fabio Paternò, Manuela Vinci, Marco Manca, and Nicola Iannuzzi. 2025. How an LLM Can Improve Automatic Web Accessibility Validation?

[6] Tommaso Calò, Alexandra-Elena Gurita, Luigi De Russis. Measuring the Semantic Accessibility Gap in LLM-Generated WebUIs. Conference on Human Factors in Computing Systems (CHI2026)

[7] Janghan Yoon, Jaegwan Cho, Junhyeok Kim, Jiwan Chung, Jaehyun Jeon, and Youngjae Yu. 2025. A11YN: Aligning LLMs for Accessible Web UI Code Generation. Preprint.

[8] Hyunjae Suh, Mahan Tafreshipour, Sam Malek, and Iftekhar Ahmed. 2025. Human or LLM? A Comparative Study on Accessible Code Generation Capability. arXiv:2503.15885 [cs.SE]

[9] Nadeen Fathallah, Daniel Hernández, and Steffen Staab. 2025. AccessGuru: Leveraging LLMs to Detect and Correct Web Accessibility Violations in HTML Code. In Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS'25). ACM. doi:10.1145/3663547.3746360

Question Time

Thank you for your attention.

Questions?